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Computational neural network in melanocytic lesions diagnosis: artificial intelligence to improve diagnosis in dermatology?

Diagnosis in dermatology is largely based on contextual factors going far beyond the visual and dermoscopic inspection of a lesion. Diagnostic tools such as the different types of dermoscopy, confocal microscopy and optical coherence tomography (OCT) are available and all of these have shown their importance in improving the dermatologist's ability, especially in the diagnosis of skin cancer. Their use, however, remains limited and time consuming, and optimizing their practice appears to be difficult, requiring extensive pre-processing, lesion segmentation and extraction of domain-specific visual features before classification. Over the last two decades, image recognition has been a matter of interest in a large part of our society and in industry, leading to the development of several techniques such as convolutional processing combined with artificial intelligence or neural networks (CNN/ANN). The aim of the present manuscript is to provide a short overview of the most recent data about CNN in the field of dermatology, mainly in skin cancer detection and its diagnosis.

Key words: artificial intelligence, computational neural network (CNN)

Introduction

Diagnosis in dermatology is largely based on contextual factors going beyond the visual and dermoscopic inspection of a lesion. Therefore, diagnostic ability depends on the experience and training of the clinician. To improve these features, there is a crucial need for increasing the sensitivity to and specificity of these classical diagnostic procedure steps in order to provide the dermatologists with new high performance diagnostic and detection tools, especially in the context of an increasing number of different types of skin cancers.

Currently, the use of dermoscopy is widely used and recognized for skin cancer diagnosis [1]. Technological evolution has led to the development of other diagnostic tools, providing the ability to explore the skin *in vivo* in depth and at a higher resolution. Among these, reflectance confocal microscopy improves diagnostic accuracy, especially in difficult-to-diagnose lesions, after dermoscopic examination. In this way, the integration of these technologies leads to a remarkable reduction of unnecessary excision, thereby maintaining excellent sensitivity [2]. Furthermore, optical coherence tomography has shown its importance to improve the dermatologist's ability, especially in the diagnosis and profiling of non-melanoma skin cancer [3-5]. However, its use relies on clinical experience and specific knowledge. Recently, attempts were made to facilitate the use of these technologies via the integration of automatic software, but this is limited due to the need for extensive pre-processing, lesion segmentation and extraction of domain-specific visual features before classification.

Over the last two decades, image recognition has been a matter of interest in industry; areas of interest for convolutional or artificial neural networks (CNN/ANN) cover, for example, automatic car driving and recognition of emergency situations recorded by street surveillance cameras. In 1998, CNN was used to recognize cursive numbers for the first time and also showed its usefulness in object and document recognition [6]. This learning process is called "deep learning" and *figure 1* provides an example of a deep learning process.

Since then, several segmentation methods have been developed. Chen *et al*. were among the first to introduce an image segmentation method via deep convolutional networks called DeepLab, while Ronneberger *et al*. built a new convolutional segmentation method called U-Net, which was an amendment and extension of the fully convolutional network (FCN) architecture [7, 8]. Long *et al*. developed the first end-to-end pixel-wise semantic segmentation technique called a FCN. Noh *et al*. developed a deep learning segmentation method called a deconvolution network (DeconvNet) as an extension of FCN, while Badrinarayanan *et al*. described a deep convolutional encoder-decoder segmentation method called SegNet [7, 9-11]. During the object classification competition in 2015, the CNN type developed by LeCun *et al*. showed an error rate of 3.6%, significantly beating the human participants [6, 12].

The aim of the present manuscript is to provide a short overview of the most recent data about CNN in the field

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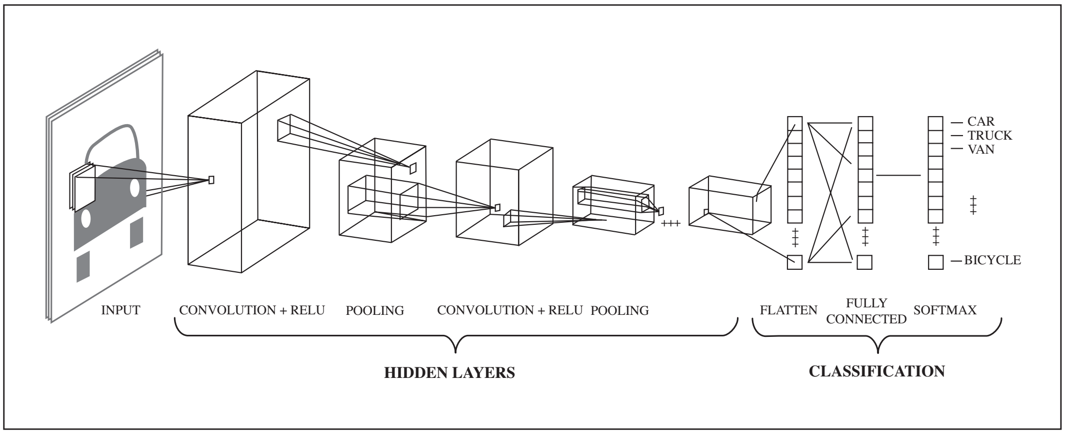

Figure 1. Example of deep learning [23].

of dermatology, mainly in skin cancer detection and its diagnosis.

Methods

The authors retrieved the most recent data published in literature regarding the impact of CNN and artificial intelligence on the identification of different skin lesion types. Results were presented at the World Rendezvous on Dermatology, on November 2nd, 2018 in Rio de Janeiro, Brazil.

Results

Neural networks in medicine

Neural networks (NN) are a form of computer software that has been subject of renewed research interest over the last decades. A NN is a distributed network of computing elements that is modeled on a biological neural system and may be implemented as a computer software program. It is capable of identifying relations in input data that are not easily apparent with current common analytic techniques. The functioning NN knowledge is built on learning and experience from previous input data. On the basis of this priorly acquired knowledge, NN can predict relations found in newly presented data sets [13]. Although NN have already been used extensively to resolve problems in engineering, they have only recently been applied to medical issues.

CNN in dermatology

In 2017, Esteva *et al*. from Stanford, California, performed an innovative study based on the development of CNN described here above. They used the GoogleNet Inception v3 CNN architecture 9 software that was pre-trained on approximately 1.28 million images. The authors "trained" this software again on a suitable dataset from their university, using a set of more than 129,450 high quality skin images. Several validation steps comparing this analysis with clinical and pathological real diagnosis were then needed. Following this, "submitted" digital images were deconstructed down to the pixel level and the above trained software "analysed" their pattern and assigned its diagnosis [14]. CNN performance was tested against 21 board-certified dermatologists on biopsy-proven clinical images with two critical binary classification use cases: keratinocyte carcinomas *versus* benign seborrheic keratoses and malignant melanomas *versus* benign nevi. The sensitivity and the specificity as well as the ROC curve (Receiver Operability Curve) were tested for CNN as well as for 21 board certified dermatologists from the USA. As a result, CNN achieved performance on par with nearly all tested clinicians across both tasks, demonstrating artificial intelligence capable of classifying skin cancer with a level of competence comparable to the median level of dermatologists.

More recently, another study compared the capacity of deep-learning CNN with an international group of 58 dermatologists from 17 countries, including 30 experts with more than five years of dermoscopic experience. Analysis was carried out using either level-I, which consisted in clinical images and dermoscopy only, or level-II, which combined previous information with age, sex and body sites [15]. A 300-image test-set was used, including 20% melanoma lesions (*in situ* and invasive) of all body sites and of all frequent histotypes, and 80% benign melanocytic nevi of different subtypes and body sites, including the so-called "melanoma simulators". The adequately trained deep learning CNN was capable of a highly accurate diagnostic classification of dermoscopic images of melanocytic origin. In conjunction with results from the reader study level-I and – II, the CNN's diagnostic performance (using a "ROC", the receiver operating curve that combines specificity and sensitivity) appeared even slightly superior to most, but not all, dermatologists. A total of 22.4% of the dermatologists showed a higher ROC curve than the CNN [16]. Importantly, these studies did not compare decision-making results, but only the diagnosis proposed. It is well-known that the important issue is the final decision, which can differ from the clinical proposal. Therefore, the best solution could be to compare the CNN *versus* the clinical decision. These studies did not include other types of pigmented lesion, nor amelanotic melanoma, and no follow-up was proposed.

As acral melanoma is the most frequent type of melanoma in Asia, Yu *et al*. from South Korea used a similar type of CNN evaluation of such lesions [16]. The authors analyzed 724 dermoscopy images comprising acral melanoma (350 images from 81 patients) and benign nevi (374 images from 194 patients), confirmed by histopathological examination. The accuracy (percentage of true positive and true negative from all images) of their CNN was higher than the two non-expert evaluations. Moreover, CNN showed area-under-the-curve (for the ROC curve) and Youden's index scores similar to that of the two experts.

A group of Chinese dermatologists assessed the capacity of deep-learning CNN trained with 19,398 images to classify the clinical images of 12 skin diseases: basal cell carcinoma, squamous cell carcinoma, intraepithelial carcinoma, actinic keratosis, seborrheic keratosis, malignant melanoma, melanocytic nevus, lentigo, pyogenic granuloma, haemangioma, dermatofibroma, and wart [17]. The capacity of the tested algorithm performance was comparable to that of 16 dermatologists for melanoma, basal cell and squamous cell carcinomas. However, additional images with a broader range of ages and ethnicities should be collected to improve the performance of CNN. Bi *et al*. proposed a cascaded multi-stage FCN followed by the parallel integration (PI) method to segment lesions from dermoscopy images and Yuan *et al*. presented a skin lesion segmentation using deep convolutional-deconvolutional neural networks (CDNN) [18, 19].

Skin classification platforms and AI models in dermatology

In their paper, Han *et al*. indicated that they had developed "a user-friendly, PC- and smartphone-based automatic skin disease classification test platform", available on https://doc.modelderm.com/, using their model [17]. The platform was developed to allow the algorithm to be tested by any interested clinician or researcher. Two models are said to be currently available for the dermatologist: Model Dermatology (http://modelderm.com) and Model Melanoma (http://melanoma.modelderm.com), which are trained with 220,680 images (174 different disease classes).

Recently, a study aimed to discover whether an artificial neural network would be able to "predict" a personal non-melanoma skin cancer (NMSC) risk factor, independently from any image [20]. This was an assay to test whether software can assign a risk factor to individuals using a questionnaire. The neural network used was trained and validated using 2056 NMSC and 460574 non-cancer cases. A total of 13 parameters were extracted, including: gender, age, BMI, diabetic status, smoking status, emphysema, asthma, race, hypertension, heart disease, vigorous exercise habits and history of stroke. The authors showed a high sensitivity and specificity, even in the absence of known UVR exposure and family history. These results indicate that the chosen model was robust enough to make predictions and to suggest novel associations and potential predictive parameters of NMSC.

Deep learning software tools in dermatology

Currently, there are several deep learning software tools under evaluation for dermatology purposes. The Berkeley Vision and Learning Center has developed CAFFE (http://www.caffe.berkelyvision.org), the Apache Software foundation has developed TORCH (http://www.torch.ch), Microsoft Research AI has prepared a cognitive toolkit named CNTH, available on (https://www.github.com/Microsoft/CNTK) and Google Brain has developed Tensorflow (https://www.tensorflow.org).

Gaps of CNN

Despite the lack of clinical studies confirming the benefit of CNN through well-designed investigations, including large series comparing decision-making steps, the use of CNN in dermatology will probably provide an undeniable tool for dermatologists in the near future and, therefore, a benefit for our patients' care. Use of CNN by dermatologists may reduce the morbidity of disease outcomes because skin diseases will be diagnosed earlier and unnecessary procedures may be avoided, thereby also reducing costs, an important factor which must not be neglected.

Other advantages may include screening for individuals requiring more urgent attention, or screening for a targeted surveillance of patients at risk, using high-resolution body imaging tools integrating an AI capacity. It does not perform any palpation, which may help clinicians and may serve as a decision support tool for the physician.

However, to date, this latter benefit still needs to be increased as no data exists concerning the success of AI in diagnosing atypical melanoma lesions, for example, on the scalp or on the mucosae. Acral surfaces have not yet been carefully evaluated, and factors such as hair, blood vessels, ruler marks, air bubbles and color illumination add obstacles within the segmentation task. In addition, amelanotic melanomas are not covered by these tools.

Moreover, a certain number of issues still need to be addressed; among those is the risk of "deskilling" the dermatologist or defining who will take responsibility for the diagnosis: the physician or the company which developed the tool? How will specificity be adjusted and what should be done regarding the recognition of amelanotic lesions?

Conclusion

Highly accurate classification of biomedical images is an essential feature in the clinical diagnosis of numerous medical diseases identified from those images, including radiology and histopathology [21]. In dermatology, several non-invasive devices such as confocal microscopy or OCT already provide convenient ways to improve the diagnosis of skin diseases, especially melanoma [16, 22]. These need training and could eventually themselves be "analyzed" by CNN in the future. Therefore, automated diagnostic systems using a CNN may improve the management, particularly when using newly developed devices [21].

Even though much research and development still has to be carried out, CNN as a form of AI has been shown to be able to achieve performance on par with clinicians across both tasks, demonstrating AI as being capable of classifying skin cancer lesions with a level of competence comparable to that of dermatologists [14-16, 20]. Even so, a significant number of clinicians remain superior in their diagnostic abilities.

In the future, CNN and AI may provide the dermatologist with an additional armamentarium of diagnosis and examination tools, allowing for even more accurate and patient- and disease-oriented clinical evaluations. Equipped with deep neural networks, mobile devices can potentially extend the reach of dermatologists outside the clinic and can, therefore, potentially provide low-cost universal access to vital diagnostic care. ■

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